IC-307D DATA STRUCTURES LAB: COMPLEXITY ANALYSIS REPORT

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1. Mathematical Foundations & Asymptotic Notations

Asymptotic analysis quantifies algorithm efficiency as input size n approaches infinity, abstracting hardware differences:

• **Big-O Notation (O):** Tight asymptotic upper bound. f(n) = O(g(n)) iff exists c > 0, n0 >= 0 such that 0 <= f(n) <= c*g(n) for all n >= n0.

• **Big-Omega Notation (Omega):** Asymptotic lower bound. f(n) = Omega(g(n)) iff exists c > 0, n0 >= 0 such that 0 <= c*g(n) <= f(n) for all n >= n0.

• **Big-Theta Notation (Theta):** Asymptotically tight bound. f(n) = Theta(g(n)) iff f(n) = O(g(n)) and f(n) = Omega(g(n)).

2. Master Theorem for Divide-and-Conquer Recurrences

For recurrences of form T(n) = a*T(n/b) + f(n) where a >= 1, b > 1:

• **Merge Sort:** T(n) = 2*T(n/2) + Theta(n). Here a=2, b=2, log_b(a) = 1. Since f(n) = Theta(n^1), Case 2 applies: **T(n) = Theta(n log n)**.

• **Binary Search:** T(n) = 1*T(n/2) + Theta(1). Here a=1, b=2, log_2(1) = 0. Since f(n) = Theta(n^0) = Theta(1), Case 2 applies: **T(n) = Theta(log n)**.

• **Quick Sort (Balanced Pivot):** T(n) = 2*T(n/2) + Theta(n) => **Theta(n log n)**. Worst case unbalanced: T(n) = T(n-1) + Theta(n) => **Theta(n^2)**.

3. Algorithm Complexity Matrix

Algorithm	Best Time	Avg Time	Worst Time	Aux Space	Mathematical Recurrence
Bubble Sort	Omega(n)	Theta(n^2)	O(n^2)	O(1)	T(n) = T(n-1) + O(n)
Selection Sort	Omega(n^2)	Theta(n^2)	O(n^2)	O(1)	T(n) = T(n-1) + (n-1)
Insertion Sort	Omega(n)	Theta(n^2)	O(n^2)	O(1)	T(n) = T(n-1) + O(n)
Merge Sort	Omega(n log n)	Theta(n log n)	O(n log n)	O(n)	T(n) = 2T(n/2) + O(n)
Quick Sort	Omega(n log n)	Theta(n log n)	O(n^2)	O(log n)	T(n) = T(k) + T(n-k-1) + O(n)
Heap Sort	Omega(n log n)	Theta(n log n)	O(n log n)	O(1)	BuildHeap O(n) + n Heapify O(log n)
Radix Sort	Omega(d(n+b))	Theta(d(n+b))	O(d(n+b))	O(n+b)	d passes of O(n + b)
Binary Search	Omega(1)	Theta(log n)	O(log n)	O(1)	T(n) = T(n/2) + O(1)

4. Empirical Benchmark Visualizations

